

Lecture 7: Newton's Method in Practice; Damping

Math 6610

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Today's Goals

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- Finish it: convergence, zero, quadratic rate
- Use it: error estimate, stopping rule, monotonicity test, damping

Recap: The Convergence Theorem

Theorem (Lecture 6). $D \subset \mathbb{R}^n$ convex, $f : D \rightarrow \mathbb{R}^n$ differentiable, $Df(x^0)$ nonsingular, and

$$\|Df(x^0)^{-1}(Df(y) - Df(x))\| \leq \gamma \|y - x\| \quad (x, y \in D), \quad \|Df(x^0)^{-1}f(x^0)\| \leq \alpha,$$
$$h := \alpha\gamma < \frac{1}{2}, \quad \overline{S_r(x^0)} \subset D, \quad r = \frac{1 - \sqrt{1 - 2h}}{\gamma}.$$

Then: (1) every x^k exists, lies in $S_r(x^0)$, $Df(x^k)$ nonsingular; (2) $x^k \rightarrow x^*$ quadratically, $f(x^*) = 0$; (3) x^* is the only zero in $D \cap S_R(x^0)$, $R = \frac{1 + \sqrt{1 - 2h}}{\gamma}$.

Remark. Everything is measured at x^0 . Nothing refers to the unknown x^* – that is what makes the theorem usable.

What the Symbols Mean

- α : length of the first Newton step, $\|x^1 - x^0\| = \|Df(x^0)^{-1}f(x^0)\| \leq \alpha$. *How far we still have to go.*
- γ : how fast Df changes. *How curved f is.* ($\gamma \rightarrow 0$: f affine, Newton finishes in one step.)
- $h = \alpha\gamma$: distance $\times$ curvature. *Does the tangent line at x^0 still describe f where we are going?*
- r : the iterates never travel farther than this. Since $h \leq 1 - \sqrt{1 - 2h} \leq 2h$,

$$\alpha \leq r \leq 2\alpha :$$

everything happens within two first-steps of x^0 .

Remark. $h < \frac{1}{2}$ is the whole hypothesis: *not too far $\times$ not too curved.*

Idea of the Proof (1/3): What Must Be Controlled

Newton can fail three ways: leave D ; hit a singular $Df(x^k)$; stop shrinking.
Step k exists only if step $k - 1$ worked $\Rightarrow$ induction on k .

So recompute the theorem's two numbers at *every* iterate:

	at x^0	at x^k
curvature	γ	γ_k (grows slowly)
step left	α	h_k/γ_k (shrinks fast)
product	$h < \frac{1}{2}$	$h_k < \frac{1}{2}$ (stays small)

Remark. γ_k grows because $Df(x^k)^{-1}$ may be larger than $Df(x^0)^{-1}$; it stays below $\gamma/\sqrt{1-2h}$. Recursions: Lecture 6 notes.

Idea of the Proof (2/3): The Worst-Case Parabola

Which f is hardest, given only α and γ ? The one that curves as much as allowed:

$$p(t) = \frac{\gamma}{2}t^2 - t + \alpha, \quad p(t) = 0 \iff t = \frac{1 \pm \sqrt{1 - 2h}}{\gamma} \in \{r, R\}.$$

r and R are its two zeros. They exist iff $1 - 2h > 0$: $h < \frac{1}{2}$ *says the worst case still has a solution.*

Run Newton on p from $t_0 = 0$: $t_k \nearrow r$. The proof shows

$$\|x^{k+1} - x^k\| \leq t_{k+1} - t_k \quad (\text{every real step} \leq \text{the matching scalar step}).$$

Remark ($h = 0.3, \gamma = 1$). $t_k = 0, 0.300, 0.3643, 0.36754, 0.367544 \rightarrow r$. Total travel $\leq r$, so the iterates stay in the ball; and they converge at least as fast as this.

Idea of the Proof (3/3): Why Steps Square

Write $\Delta := x^k - x^{k-1}$. The previous step was chosen to kill the tangent model: $f(x^{k-1}) + Df(x^{k-1})\Delta = 0$.

So $f(x^k)$ is exactly *the error of that tangent model* – and the mean value theorem writes it as a difference of Jacobians:

$$f(x^k) = \int_0^1 \underbrace{(Df(x^{k-1} + t\Delta) - Df(x^{k-1}))}_{\leq \gamma_k t \|\Delta\| \text{ by Lipschitz}} \Delta dt \implies \|x^{k+1} - x^k\| \leq \frac{\gamma_k}{2} \|\Delta\|^2.$$

Remark. *New step* $\leq \text{const} \times (\text{old step})^2$. One factor $\|\Delta\|$ because the Jacobians are compared over a distance $\|\Delta\|$, one because that difference multiplies Δ . This is the whole mechanism.

What the Induction Delivers

Proposition (Lecture 6 notes). For every $k = 0, 1, 2, \dots$

(a) $x^k \in S_r(x^0)$ (b) $Df(x^k)$ nonsingular

(c) $\|Df(x^k)^{-1}(Df(y) - Df(x))\| \leq \gamma_k \|y - x\|$, $\gamma_k < \frac{\gamma}{\sqrt{1 - 2h}}$

(d) $\|x^{k+1} - x^k\| \leq \frac{h_k}{\gamma_k} \leq \frac{\alpha}{2^k}$

(a), (b) are Part 1. Take (a)–(d) as given; two short steps now give Part 2. Part 3 (uniqueness): notes.

Finishing the Proof (1/2): Convergence to a Zero

Proof. Cauchy. Steps shrink geometrically by (d), so for $m > k$ the tail sums:

$$\|x^m - x^k\| \leq \frac{\alpha}{2^{m-1}} + \cdots + \frac{\alpha}{2^k} < \frac{\alpha}{2^k} \underbrace{\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right)}_{=2} = \frac{\alpha}{2^{k-1}} \rightarrow 0,$$

independently of m . $\mathbb{R}^n$ is complete: $x^* := \lim x^k$ exists, and $x^* \in \overline{S_r(x^0)}$ by (a).

The limit is a zero. Rearranging the Newton equation, $f(x^k) = -Df(x^k)(x^{k+1} - x^k)$. The steps go to 0 and $\|Df\|$ is bounded on the closed ball by some M , so

$$\|f(x^k)\| \leq M \frac{\alpha}{2^k} \rightarrow 0 \quad \implies \quad f(x^*) = \lim f(x^k) = 0.$$

(continued)

Finishing the Proof (2/2): The Rate Is Quadratic

Proof (continued). Same trick as before, now comparing x^k with x^* instead of with x^{k-1} . Subtract x^* from $x^{k+1} = x^k - Df(x^k)^{-1}f(x^k)$ and use $f(x^*) = 0$:

$$x^{k+1} - x^* = \int_0^1 Df(x^k)^{-1} \left(Df(x^k) - Df(x^* + t(x^k - x^*)) \right) (x^k - x^*) dt.$$

Again a difference of Jacobians, over the distance $(1-t) \|x^k - x^*\|$. By (c),

$$\|x^{k+1} - x^*\| \leq \gamma_k \|x^k - x^*\|^2 \int_0^1 (1-t) dt = \frac{\gamma_k}{2} \|x^k - x^*\|^2 \leq \underbrace{\frac{\gamma}{2\sqrt{1-2h}}}_{=:C} \|x^k - x^*\|^2.$$



What Quadratic Convergence Looks Like

$C \|x^{k+1} - x^*\| \leq (C \|x^k - x^*\|)^2$: once the scaled error is below 1, it squares every step.

k	0	1	2	3	4	5
error	10^{-1}	10^{-2}	10^{-4}	10^{-8}	10^{-16}	—
correct digits	1	2	4	8	16	(machine)

Remark. Four or five iterations from a decent guess – then double precision runs out. A method that needs 20 iterations is not Newton converging; it is Newton *not yet* converging, i.e. h is not small yet.

The Step Tells You the Error

We never know $\|x^k - x^*\|$. But $x^k - x^* = -\Delta x^k + (x^{k+1} - x^*)$, and the last term is quadratically small:

$$\left| \|x^k - x^*\| - \|\Delta x^k\| \right| \leq C \|x^k - x^*\|^2.$$

Remark (Stopping rule). Near x^* the computed step $\|\Delta x^k\|$ is the error. Stop when

$$\|\Delta x^k\| \leq \text{tol}.$$

Free (already computed), and affine-invariant: Δx^k does not change under $f \mapsto Mf$ (Lecture 5), whereas $\|f(x^k)\|$ does.

The Wrong Test and the Right One

Far from x^* , h is not small and the theorem says nothing. We must *check* each step.

Wrong: $\|f(x^{k+1})\| \leq \theta \|f(x^k)\|$, $\theta < 1$. Rescale $f \mapsto Mf$: iterates identical, test outcome can flip. Not affine-invariant.

Right: measure the residual the way Newton does, through $Df(x^k)^{-1}$:

$$\|Df(x^k)^{-1}f(x^{k+1})\| \leq \theta \underbrace{\|Df(x^k)^{-1}f(x^k)\|}_{= \|\Delta x^k\|}, \quad (MDf)^{-1}Mf = Df^{-1}f.$$

Both sides survive rescaling. *Compare like with like: steps, not residuals.*

The Monotonicity Test

Definition (Simplified Newton correction). $\overline{\Delta x}^{k+1}$ solves $Df(x^k) \overline{\Delta x}^{k+1} = -f(x^{k+1})$: the *new* residual with the *old* Jacobian. Then $\|Df(x^k)^{-1}f(x^{k+1})\| = \|\overline{\Delta x}^{k+1}\|$.

Remark (Test).

- Solve, and set $x^{k+1} = x^k + \Delta x^k$:

$$Df(x^k) \Delta x^k = -f(x^k) \quad (1)$$

- $\overline{\Delta x}^{k+1}$ uses the *same* matrix as (1) $\Rightarrow$ reuse its LU factors: $O(n^2)$ extra, not $O(n^3)$.
- Accept if $\|\overline{\Delta x}^{k+1}\| \leq \theta \|\Delta x^k\|$, typically $\theta = \frac{1}{2}$.
- Otherwise the full step is not trusted $\Rightarrow$ damping.

Newton's Method with Damping

Remark (*Standard vs. damped*).

$$x^{k+1} = x^k + \Delta x^k \quad \longrightarrow \quad x^{k+1}(\lambda_k) = x^k + \lambda_k \Delta x^k, \quad 0 < \lambda_k \leq 1.$$

Remark (*Why*). Δx^k is the right *direction* – it comes from the tangent model at x^k . Far away, that model is only trustworthy nearby (h large), so we believe the direction but not the full length. Damping walks part of the way, re-linearizes, and tries again. Near x^* we must return to $\lambda_k = 1$, or we lose the quadratic rate.

Remark. The test says whether a step is acceptable; damping supplies shorter candidates when it is not. *How should λ_k be chosen?* – next lecture.

Looking Ahead

- **Next lecture:** Damped Newton; Introduction to Conditioning